Q1. JEE Main 2026 (21 January Shift 2)

If $\left(\frac{1}{{}^{15}C_0}+\frac{1}{{}^{15}C_1}\right)\left(\frac{1}{{}^{15}C_1}+\frac{1}{{}^{15}C_2}\right)\cdots\left(\frac{1}{{}^{15}C_{12}}+\frac{1}{{}^{15}C_{13}}\right)=\frac{\alpha^{13}}{{}^{14}C_0{}^{14}C_1\cdots{}^{14}C_{12}}$, then 30α is equal to ____.

Q2. JEE Main 2026 (23 January Shift 1)

The value of $\frac{{}^{100}C_{50}}{51}+\frac{{}^{100}C_{51}}{52}+\ldots+\frac{{}^{100}C_{100}}{101}$ is :

(1) $\frac{2^{101}}{100}$ (2) $\frac{2^{100}}{100}$ (3) $\frac{2^{101}}{101}$ (4) $\frac{2^{100}}{101}$

Q3. JEE Main 2026 (23 January Shift 1)

The sum of all possible values of $n \in \mathbf{N}$, so that the coefficients of x, x^2 and x^3 in the expansion of $\left(1+x^2\right)^2(1+x)^n$, are in arithmetic progression is :

(1) 12 (2) 7 (3) 3 (4) 9

Q4. JEE Main 2026 (28 January Shift 2)

Given below are two statements :

Statement I: $25^{13}+20^{13}+8^{13}+3^{13}$ is divisible by 7.

Statement II: The integral part of $(7+4\sqrt{3})^{25}$ is an odd number.

In the light of the above statements, choose the correct answer from the options given below :

(1) Statement I is true but Statement II is false

(2) Both Statement I and Statement II are true

(3) Statement I is false but Statement II is true

(4) Both Statement I and Statement II are false

Q5. JEE Main 2026 (28 January Shift 2)

The sum of the coefficients of x^{499} and x^{500} in $(1+x)^{1000}+x(1+x)^{999}+x^2(1+x)^{998}+\ldots+x^{1000}$ is :

(1) ${}^{1002}C_{500}$ (2) ${}^{1002}C_{501}$

(3) ${}^{1001}C_{501}$ (4) ${}^{1000}C_{501}$

Q6. JEE Main 2026 (22 January Shift 2)

Let C_r denote the coefficient of x^r in the binomial expansion of $(1+x)^n, n \in N, 0 \leq r \leq n$. If $P_n=C_0-C_1+\frac{2^2}{3}C_2-\frac{2^3}{4}C_3+\ldots\ldots+\frac{(-2)^n}{n+1}C_n$, then the value of $\sum_{n=1}^{25}\frac{1}{P_{2n}}$ equals.

(1) 650 (2) 675 (3) 525 (4) 580

Q7. JEE Main 2026 (22 January Shift 1)

The coefficient of x^{48} in $(1+x)+2(1+x)^2+3(1+x)^3+\ldots+100(1+x)^{100}$ is equal to

(1) $100\cdot{}^{100}C_{49}-{}^{100}C_{48}$ (2) ${}^{100}C_{50}+{}^{101}C_{49}$

(3) ${}^{100\cdot{}^{101}C_{49}-{}^{101}C_{50}}$ (4) $100\cdot{}^{100}C_{49}-{}^{100}C_{50}$

Q8. JEE Main 2026 (21 January Shift 1)

If the coefficient of x in the expansion of $\left(ax^2+bx+c\right)(1-2x)^{26}$ is -56 and the coefficients of x^2 and x^3 are both zero, then a + b + c is equal to :

(1) 1500 (2) 1300 (3) 1403 (4) 1483

Q9. JEE Main 2026 (24 January Shift 1)

Let S $= \frac{1}{25!}+\frac{1}{3!23!}+\frac{1}{5!21!}+\ldots$ up to 13 terms. If 13 S $= \frac{2^k}{n!}, k \in$ N, then $n+k$ is equal to

(1) 50 (2) 49 (3) 52 (4) 51

ANSWERS AND SOLUTIONS

1. 32 **2.** (4) **3.** (4) **4.** (2) **5.** (1) **6.** (2) **7.** (3) **8.** (3)
9. (2)

1. **(32)** The general term of the product is $T_r = \frac{1}{{}^{15}C_{r-1}} + \frac{1}{{}^{15}C_r}$ for $r = 1, 2, \ldots, 13$.
Using the formula ${}^{n}C_r = \frac{n}{r} \cdot {}^{n-1}C_{r-1}$, we have ${}^{15}C_r = \frac{15}{r} \cdot {}^{14}C_{r-1}$ and
${}^{15}C_{r-1} = \frac{15}{15-(r-1)} \cdot {}^{14}C_{r-1} = \frac{15}{16-r} \cdot {}^{14}C_{r-1}$.
Substituting these into T_r:
$T_r = \frac{16-r}{15 \cdot {}^{14}C_{r-1}} + \frac{r}{15 \cdot {}^{14}C_{r-1}} = \frac{16-r+r}{15 \cdot {}^{14}C_{r-1}} = \frac{16}{15 \cdot {}^{14}C_{r-1}}$.
The product is $\prod_{r=1}^{13} T_r = \prod_{r=1}^{13} \frac{16}{15 \cdot {}^{14}C_{r-1}} = \frac{16^{13}}{15^{13} \cdot \prod_{r=1}^{13} {}^{14}C_{r-1}}$.
The denominator of the product is $15^{13} \cdot ({}^{14}C_0 \cdot {}^{14}C_1 \ldots {}^{14}C_{12})$.
Comparing this with the given expression $\frac{\alpha^{13}}{{}^{14}C_0 \cdot {}^{14}C_1 \ldots {}^{14}C_{12}}$, we get:
$\alpha^{13} = \frac{16^{13}}{15^{13}} \implies \alpha = \frac{16}{15}$.
We need to find the value of 30α:
$30\alpha = 30 \times \frac{16}{15} = 2 \times 16 = 32$.

2. **(4)** Using $\frac{\binom{n}{r}}{r+1} = \frac{1}{n+1}\binom{n+1}{r+1}$:
$\sum_{r=50}^{100} \frac{\binom{100}{r}}{r+1} = \frac{1}{101} \sum_{r=50}^{100} \binom{101}{r+1} = \frac{1}{101} \sum_{k=51}^{101} \binom{101}{k}$.
Since 101 is odd, by symmetry $\sum_{k=51}^{101} \binom{101}{k} = \sum_{k=0}^{50} \binom{101}{k} = \frac{2^{101}}{2} = 2^{100}$.
Required value $= \frac{2^{100}}{101}$.

3. **(4)** $(1+x^2)^2 = 1 + 2x^2 + x^4$. Multiply with $(1+x)^n$:
Coefficient of x: n.
Coefficient of x^2: $\binom{n}{2} + 2 = \frac{n(n-1)}{2} + 2$.
Coefficient of x^3: $\binom{n}{3} + 2n = \frac{n(n-1)(n-2)}{6} + 2n$.
For AP: $2\left(\frac{n(n-1)}{2} + 2\right) = n + \frac{n(n-1)(n-2)}{6} + 2n$.
Simplifying: $n^3 - 9n^2 + 26n - 24 = 0$.
$(n-2)(n-3)(n-4) = 0 \Rightarrow n = 2, 3, 4$.
Sum $= 2 + 3 + 4 = 9$.

4. **(2)** Statement I:
$25^{13} + 3^{13} \rightarrow$ divisible by $(25+3)$
$20^{13} + 8^{13} \rightarrow$ divisible by $(20+8)$
$\therefore$ divisible by 7. TRUE
Statement II:
For $(7+4\sqrt{3})^{25}$, let $\alpha = 7 + 4\sqrt{3}$ and $\beta = 7 - 4\sqrt{3}$ (conjugate).

Then $\alpha\beta = 1$ and $\alpha + \beta = 14$.

Since $0 < \beta < 1$, we have $\alpha^{25} + \beta^{25}$ is an integer.

The fractional part of α^{25} is $1 - \beta^{25}$, making the integral part odd. TRUE.

Both Statement I and Statement II are true.

5. **(1)** $S = (1+x)^{1000} + x(1+x)^{999} + x^2(1+x)^{998} + \cdots + x^{1000}$

This is a geometric series with first term $(1+x)^{1000}$, common ratio $\frac{x}{1+x}$, and 1001 terms.

$$S = (1+x)^{1000} \cdot \frac{1 - \left(\frac{x}{1+x}\right)^{1001}}{1 - \frac{x}{1+x}}$$

$= (1+x)^{1001} - x^{1001}$

Required sum = coefficient of x^{499} + coefficient of x^{500} in $(1+x)^{1001} - x^{1001}$

$= {}^{1001}C_{499} + {}^{1001}C_{500} = {}^{1002}C_{500}$

6. **(2)** Using the identity $\frac{1}{r+1}\binom{n}{r} = \frac{1}{n+1}\binom{n+1}{r+1}$,

we get $P_n = \sum_{r=0}^{n} \frac{(-2)^r}{r+1} C_r = \frac{1}{n+1}\sum_{s=1}^{n+1}(-2)^s\binom{n+1}{s}$.

Since $(1-2)^{n+1} = (-1)^{n+1} = \sum_{s=0}^{n+1}(-2)^s\binom{n+1}{s}$,

we have $\sum_{s=1}^{n+1}(-2)^s\binom{n+1}{s} = (-1)^{n+1} - 1$.

For even $n = 2k$: $P_{2k} = \frac{1-(-1)}{2(2k+1)} = \frac{1}{2k+1}$.

Thus $\frac{1}{P_{2n}} = 2n+1$, and $\sum_{n=1}^{25}(2n+1) = 2 \cdot \frac{25 \cdot 26}{2} + 25 = 675$.

7. **(3)** Let $1 + x = r$

$\therefore S = 1 \cdot r + 2 \cdot r^2 + 3 \cdot r^3 + \ldots + 100r^{100}$ .....(1) (AGP)

$rS = 1 \cdot r^2 + 2 \cdot r^3 + \ldots + 99r^{100} + 100r^{101}$ .....(2)

(1) − (2) gives

$$S = -\frac{(1+x)^{101}}{x^2} + \frac{1}{x^2} + \frac{100(1+x)^{101}}{x}$$

$\therefore$ coefficient x^{48} in S

$= -$ coefficient of x^{48} in $\frac{(1+x)^{101}}{x^2} + 100\cdot$ Coefficient of x^{48} in $\frac{(1+x)^{101}}{x}$

$= 100 \cdot {}^{101}C_{49} - {}^{101}C_{50}$

8. **(3)** In $(1-2x)^{26}$: $T_0 = 1$, $T_1 = -52$, $T_2 = 1300$, $T_3 = -20800$.

Coefficient of x: $b - 52c = -56$ ...(1)

Coefficient of x^2: $a - 52b + 1300c = 0$ ...(2)

Coefficient of x^3: $-52a + 1300b - 20800c = 0$ ...(3)

From (1): $b = 52c - 56$. Substituting in (2): $a = 1404c - 2912$.

Substituting in (3): $-26208c + 78624 = 0 \Rightarrow c = 3$.

Thus $b = 100$, $a = 1300$.

$a + b + c = 1300 + 100 + 3 = 1403$.

9. **(2)** The general term is $\frac{1}{(2r-1)!(27-2r)!}$ for $r = 1$ to 13.

Multiply and divide by 26!: $\frac{1}{(2r-1)!(27-2r)!} = \frac{1}{26!}\binom{26}{2r-1}$

$S = \frac{1}{26!}\sum_{r=1}^{13}\binom{26}{2r-1} = \frac{1}{26!}\left[\binom{26}{1} + \binom{26}{3} + \cdots + \binom{26}{25}\right]$

Sum of odd-indexed binomial coefficients: $\sum\binom{26}{\text{odd}} = 2^{25}$

$S = \frac{2^{25}}{26!}$

$13S = \frac{13\cdot 2^{25}}{26!} = \frac{13\cdot 2^{25}}{26\cdot 25!} = \frac{2^{25}}{2\cdot 25!} = \frac{2^{24}}{25!}$

So $k = 24,\ n = 25$, giving $n + k = 49$.